Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Klemens Ilse	Email	klemens.ilse@imws.fraunhofer.de
Institute Name	Fraunhofer Institute for Microstructure of Materials and Systems IMWS			
Address	Walter-Huelse-Str 1, D-06120 Halle, Germany			

2. Technology Overview

Title of Proposed Technology	*State of Health Monitoring of Electrolyzer Stacks using Magnetic Field Analysis (MFA)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Measuring Instruments*	*Safety Monitoring/Diagnosis Control Instruments*
Concept of Proposed Technology	• *Non-destructive & non-invasive* **Magnetic Field Analysis (MFA)** to "see inside" operating Electrolyzer stacks.* • *External or integrated sensors measure the magnetic field; conclusion on **the internal current distribution.*** • *Early detection of critical issues: inhomogeneous current density, inactive MEA areas, metallic debris, stray currents in the BoP.* • *Main uses: **State-of-Health monitoring**, predictive maintenance, safe restart after emergency shutdowns, and **stack design optimization**. *when measured externally*			

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Rationale for Joint R&D Project	• *Today, operators have no simple tool to assess internal current distribution and SoH in industrial stacks → decisions are made with limited insight.* • *Current status of MFA: **TRL 3** (feasibility proven on research stacks and first industrial tests); target: **TRL 6** (demonstration in relevant industrial environment).* • *Fraunhofer IMWS brings **sensors with mapping technology** and **magnetic tomography know-how.*** • *Korean SME partner **operating and ideally producing electrolyzer stacks** provides access to real plants, test stacks, operating data and integration constraints.* • *Joint R&D enables a **differentiating diagnostic feature** and **new service revenues** (health checks, troubleshooting, predictive maintenance) for the Korean partner.*
Key Words	*1. Magnetic Field Imaging for Electrolyzer Stacks* *2. State-of-Health (SoH) Monitoring & Predictive Maintenance* *3. Non-Invasive Current Distribution Diagnostics*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• *Develop **industrial MFI measurement concepts** for electrolyzer stacks:* ○ *External sensor rings or mobile scanning systems for existing stacks* ○ *Sensor-equipped bipolar plates for internal operando measurements on research/next-gen stacks* • *Identify **detectable defects and operating conditions** in industrial environments, e.g.:* ○ *Current density inhomogeneities and evolving degradation* ○ *Deactivated MEA areas, metallic debris, unintended current paths* ○ *Stray currents within the Balance of Plant (BoP)* • *Develop and optimize **data processing and visualization** to convert magnetic field maps into robust, easy-to-interpret SoH indicators (heat maps, alarms, KPIs).* • *Validate MFI under **real operating conditions** (start-up, steady-state, load changes, emergency shutdown/restart) on full-scale stacks operated by the Korean SME.* • *Prepare **integration and commercialization concepts**: diagnostic kit / integrated feature, standard operating procedures (SOPs), and a basis for joint IP and product roadmap.*			
R&D Target Specifications	• ***Defect detection capability*** ○ *Detect deactivated MEA regions of **≥ 1 cm²** and local current-density deviations of **≥ 10 %** at typical industrial current densities* ○ *Detect metallic debris or unintended current paths causing **≥ 5 %** redistribution of local current.* • ***Localization & resolution*** ○ *Localize anomalies with an axial accuracy of **better than ± one cell** (depending on stack design).* • ***Measurement speed*** ○ *Mobile scanning system: complete scan of one accessible stack side in **≤ 10 min**.* ○ *Fixed external sensor rings / internal sensor plates: data update rate **≤ 1 s** for continuous monitoring.* • ***Reliability & robustness*** ○ *Sensor system operable in industrial environment (temperature, vibration, EM noise) with **> 95 %** system uptime during field trials.* • ***Field validation*** ○ *Demonstration of MFI-based SoH monitoring on at least **one industrial stack** over **≥ 1,000 h** cumulative monitored operation at the Korean SME.* ○ *Quantified comparison vs. standard diagnostics (cell voltages, global performance): demonstration of **added diagnostic value** (earlier or more precise fault localization).*			
R&D Plan	***WP1 – Development of MFI Measurement Concepts (Year 1)*** • *Jointly define **industrial use cases and KPIs** (safety, uptime, OPEX, maintenance).* • *Collect detailed **stack design and operating data** from the Korean SME (geometry, electrical layout, BoP).* • *Design and adapt **external sensor rings**, **mobile scanning systems** and **sensor-equipped bipolar plates** to the partner's stacks (including ATEX concepts where needed).* • *Laboratory tests on **research stacks / pilot stacks** with defined artificial defects to verify detection limits and optimize sensor layouts.* ***WP2 – Validation & Correlation of Measured Data (Year 1–2)***			

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	• *Conduct **measurement campaigns** on operating stacks at the Korean SME:*
	  ○ *Start-up, steady-state at different loads, overload, shutdown, emergency stop / restart.*
	  ○ *Long-term vs. short-term, spatial (scanning) vs. local (fixed sensors), external vs. internal measurements.*
	• *Correlate MFI results with **existing monitoring data** (cell voltages, stack current, temperature, incidents) and, where available, **simulation results** (e.g. magnetic field / current distribution models).*
	• *Identify and document **detectable defect classes**, detection thresholds and blind spots.*
	
	WP3 – Limitations, Benefits, Benchmarking & Industrialization (Year 2–3)
	• ***Benchmark** MFI against conventional diagnostic methods (e.g. cell-voltage monitoring, thermography, segmented cells) to quantify added value and limitations.*
	• *Perform **cost–benefit analysis** for different deployment options:*
	  ○ *Mobile service tool vs. permanently installed diagnostic feature.*
	• *Define **standard operating procedures (SOPs)** and test protocols for:*
	  ○ *End-of-line quality control,*
	  ○ *Operando SoH monitoring and predictive maintenance,*
	  ○ *Safe recommissioning after faults or emergency shutdowns.*
	• *Design a **pre-commercial diagnostic kit or reference architecture** (hardware + software + procedures) ready for industrial integration by the Korean SME.*
	• *Prepare **joint exploitation plan** (integration into product portfolio, service offering, licensing/IP strategy) and a roadmap towards commercialization.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Technology leadership for Magnetic Field Diagnostics for electrolyzer stacks.* • *Design, construction and lab-validation of sensor systems (external sensor rings, mobile scanner, sensor-equipped bipolar plates).* • *Development and optimization of, SoH indicators and visualization tools (current-density maps, alarms, KPIs).* • *Execution of controlled lab experiments and pilot tests on research stacks; correlation with complementary diagnostics (e.g. CT, thermography, electrochemical tests).* • *Definition of test protocols, data evaluation methods and SOPs; training of Korean partner staff.* • *Joint IP generation and support for certification/safety concepts, while protecting core know-how.*
Expected R&R of Korean SME Partner	• *Provision of industrial electrolyzer stacks (design data, sample stacks) and access to operating plants for field measurements.* • *Mechanical and electrical integration of MFA sensor systems into existing stack designs and BoP, including ATEX compliance at the site.* • *Operation of test campaigns under realistic conditions (start up, nominal load, overload, shutdown) and provision of operational data (cell voltages, process data, maintenance history).* • *Feedback on industrial requirements (cost, reliability, serviceability) and co development of product concepts (diagnostic kit, integrated monitoring option, service offering).* • *Lead role in industrialization and commercialization in Korea (manufacturing, sales, customer support) based on the jointly developed technology.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Demonstration of MFI as a reliable, non-invasive diagnostic tool for full-scale electrolyzer stacks in an industrial environment (TRL increase from 3 to 6).* • *Validated sensor concepts (external rings, mobile scanner, sensor bipolar plate) with quantified detection limits for typical defect classes (e.g. deactivated MEA areas, metallic debris, stray currents).* • *Software toolkit for fast and intuitive visualization of internal current density and SoH indicators, ready for integration into plant monitoring systems.* • *Standardized test and service procedures (end-of-line test, operando SoH monitoring, recommissioning after faults) tailored to the Korean SME's product portfolio.* • *Joint IP and a pre-commercial diagnostic kit / reference architecture that the Korean SME can bring to market shortly after project completion.* ***Mid- to Long Term Industrial and Economic Impact:*** • *Higher reliability, availability and safety of industrial electrolyzers, leading to reduced unplanned downtime and lower OPEX for operators.* • *Differentiation of the Korean SME's stack products through advanced built in diagnostics and premium service offerings (predictive maintenance, fast troubleshooting).* • *Creation of new revenue streams from diagnostic services, retrofit solutions and data driven performance optimization of installed electrolyzers.* • *Strengthening of Korea's position in the global hydrogen value chain by combining domestic stack manufacturing with unique monitoring technology from the joint project.* • *Reduced technical and financial risk for large hydrogen projects, supporting faster investment decisions.*

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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